

memorizing all the labels on the training data, and that generalization happening at all requires a non-trivial explanation.

A.5 GENERALIZATION MEASURES

We believe it is useful to explore how predictive common generalization measures are of generalization on small algorithmic datasets presented in this paper. In a preliminary investigation we found that sharpness Hochreiter & Schmidhuber (1997) of the minimum found by a trained network measure seems to be predictive of generalization on one of these datasets. We trained multiple networks with different initialization seeds for a fixed number a steps on the *S5* composition objective, until approximately half of them achieved high validation accuracy. We then used the method described in Keskar et al. (2016) to calculate the sharpness approximation value, ϕ . We found that the validation accuracy and the ϕ score across our trained networks had Spearman correlation coefficient of -0.79548 (significant with $p < 0.000014$). This is suggestive that grokking may only happen after the network’s parameters are in flatter regions of the loss landscape. It would be valuable for future work to explore this hypothesis, as well as test other generalization measures.

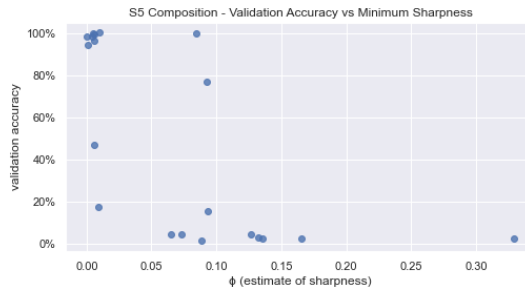


Figure 7: Networks trained on the *S5* composition objective appear to only grok in relatively flat regions of the loss landscape.